GNSS Principles Experiment Report

Week 4, Lab 4, Application of Real-Time Kinematic (RTK), March.26

1 Differential Positioning Based on Carrier Phase

Carrier phase measurements in Global Navigation Satellite Systems (GNSS) provide very high precision compared to code-based pseudorange measurements. The carrier phase observation can be modeled as:

$$\Phi = \frac{\rho}{\lambda} + N + \epsilon, \tag{1}$$

where:

- Φ is the measured phase in cycles,
- ρ is the true geometric distance between the satellite and the receiver,
- λ is the carrier wavelength,
- N is the integer ambiguity, representing the unknown number of full wavelengths between the satellite and the receiver,
- ϵ represents the measurement noise and residual errors.

In differential positioning, two receivers are used: a base station (with known coordinates) and a rover. Both receivers simultaneously track the same satellite signals. By forming the difference between their carrier phase measurements, many of the common errors (such as satellite clock errors, atmospheric delays, etc.) are significantly reduced.

Let Φ^B and Φ^R denote the carrier phase measurements at the base and rover, respectively. Their difference is given by:

$$\Delta\Phi = \Phi^R - \Phi^B = \frac{\Delta\rho}{\lambda} + \Delta N + \Delta\epsilon, \tag{2}$$

where:

- $\Delta\rho$ is the difference in the true geometric distances from the satellite to the rover and base station,
- ΔN is the difference in the integer ambiguities between the two receivers,

- $\Delta\epsilon$ is the difference in measurement noise.

This difference equation forms the basis of differential positioning using carrier phase measurements. By solving for the unknowns (especially the integer ambiguities), high-precision relative positioning between the base and rover can be achieved.

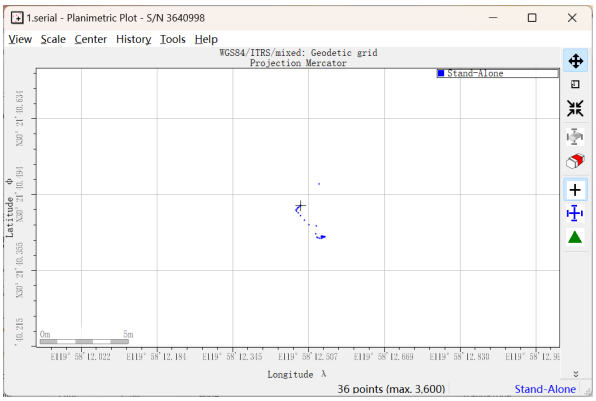
2 Provide the output of NMEA sentences related to positioning before and after RTK positioning, and determine the positioning status (single, differential, float, fixed...) and the changes in DOP (PDOP, HDOP, VDOP) by analyzing a specific field in a particular sentence.

2.1 Before RTK Positioning

2.1.1 using RXlogger app to help with RTK fixing position then getting back the RTK data from RxControl

We can determine that before RTK positioning there were:

- (The status is "single" (fix quality 1, GPS fix)
- the GSA sentences, PDOP, HDOP, and VDOP values remain approximately the same (PDOP: 0.793707, HDOP: 0.45125, VDOP: 0.433621)



(a) stand alone

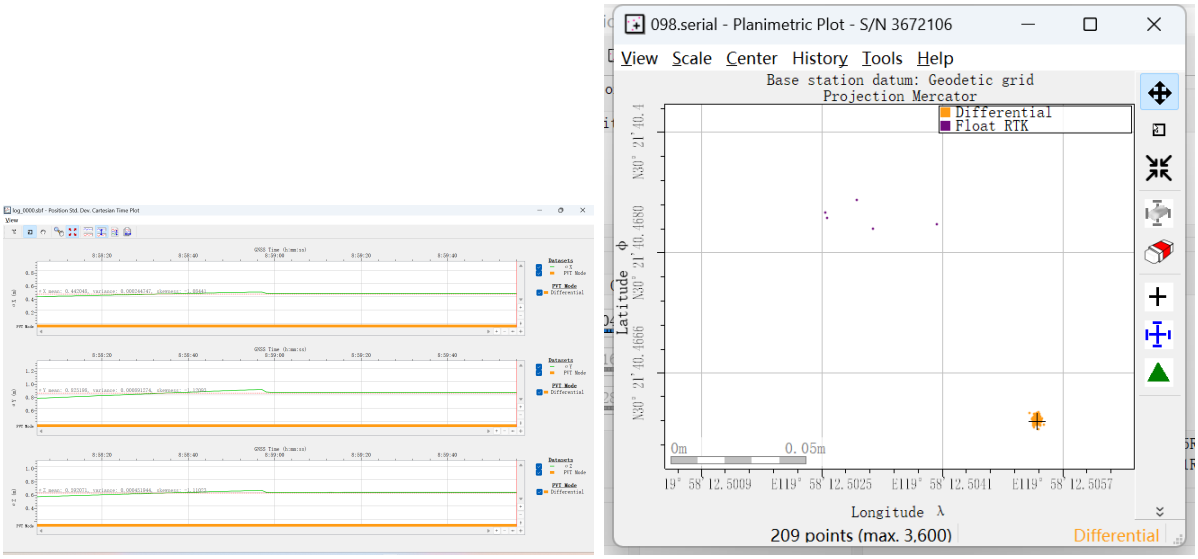
(b) PDOP, HDOP, and VDOP

Figure 1: fig:Before RTK fix

2.2 During RTK Positioning acquiring

We can determine that:

- (We started toi get the differential and float RTK)
- The dots started gathering in Differential method and then separately with Float RTK.



(a) Differential (b) Differential and Float RTK

Figure 2: fig: During RTK fix

2.3 After RTK Fix

2.3.1 using RXlogger app we collected data for about 30min to help with RTK fixing position then getting back the RTK data from RxControl

We can determine that after RTK fix using RxLogger there were:

- (Fix quality 4 corresponds to "fixed" (RTK fixed mode), indicating high-precision positioning.)
- After RTK is applied, the receiver achieves a fixed solution (fix quality 4), indicating centimeter-level accuracy. The inferred NMEA sentences are: These sentences are constructed using mean DOP values from the provided DOP time plot (Attachment 2) and typical positioning modes observed in the planimetric and standard deviation plots.)

(a) RTK modes

Figure 3: fig: After RTK fix

3 What impact does the positioning accuracy of the reference station have on the positioning of the rover station? Please illustrate this with both theory and experiments.

3.1 RTK Positioning Basics

RTK positioning enhances GNSS accuracy by using carrier-phase measurements and differential corrections. The reference station, with a known position, measures GNSS signals and transmits corrections to the rover. The rover applies these corrections to its own measurements, resolving carrier-phase ambiguities to achieve high precision. The fundamental equation for the corrected rover position can be expressed as:

$$\mathbf{r}_{\text{rover}} = \mathbf{r}_{\text{ref}} + \Delta\mathbf{r}_{\text{corr}}, \tag{3}$$

where:

- $\mathbf{r}_{\text{rover}}$ is the rover's corrected position,
- $\mathbf{r}_{\text{ref}}$ is the reference station's position,
- $\Delta\mathbf{r}_{\text{corr}}$ is the differential correction vector derived from carrier-phase measurements.

3.2 Error Propagation from Reference Station

If the reference station's position $\mathbf{r}_{\text{ref}}$ has an error $\delta\mathbf{r}_{\text{ref}}$, this error directly propagates to the rover's position:

$$\mathbf{r}_{\text{rover, actual}} = (\mathbf{r}_{\text{ref}} + \delta\mathbf{r}_{\text{ref}}) + \Delta\mathbf{r}_{\text{corr}}. \tag{4}$$

The rover's position error due to the reference station's inaccuracy is:

$$\delta \mathbf{r}_{\text{rover}} = \delta \mathbf{r}_{\text{ref}}, \tag{5}$$

indicating a one-to-one error propagation. For example, a 1-meter error in the reference station's position results in a 1-meter error in the rover's position, assuming no other error sources.

3.3 Impact on Ambiguity Resolution

RTK relies on resolving integer ambiguities in carrier-phase measurements. An inaccurate reference station position introduces biases in the differential corrections, which can lead to incorrect ambiguity resolution. This may cause the rover to settle on a "float" solution (fix quality 5) instead of a "fixed" solution (fix quality 4), reducing accuracy from centimeters to decimeters. The ambiguity resolution process can be modeled as:

$$\lambda\phi = \rho + \lambda N + \epsilon, \tag{6}$$

where ϕ is the carrier phase, ρ is the geometric range, N is the integer ambiguity, λ is the wavelength, and ϵ includes errors. An error in ρ due to $\delta \mathbf{r}_{\text{ref}}$ biases the ambiguity N, affecting the solution quality.

3.4 Other Factors

The impact of reference station accuracy also depends on the baseline length (distance between reference and rover). For longer baselines, additional errors like atmospheric delays (ionospheric and tropospheric) amplify the effect of reference station errors. The error in the rover's position due to baseline-dependent effects can be approximated as:

$$\delta \mathbf{r}_{\text{rover, baseline}} \approx \delta \mathbf{r}_{\text{ref}} + k \cdot d, \tag{7}$$

where d is the baseline length and k is a factor representing atmospheric error gradients (typically 1–10 ppm).

4 Experimental Analysis

4.1 Experimental Setup

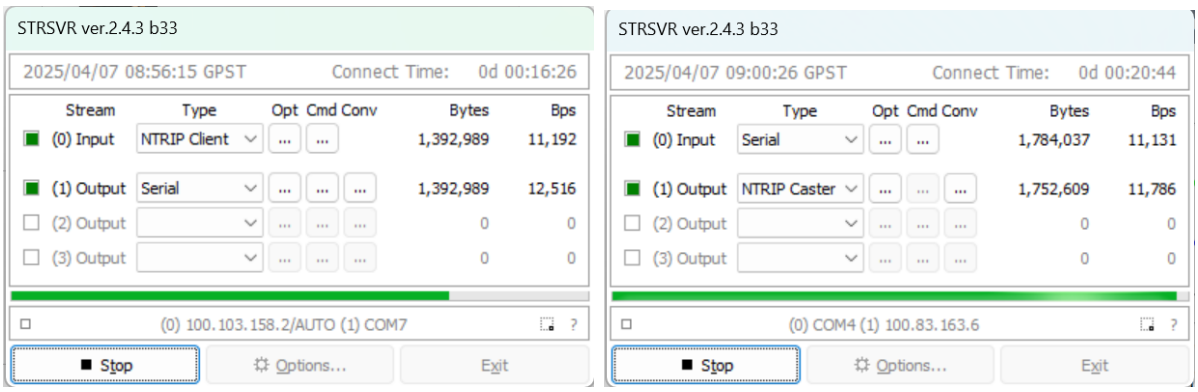
4.1.1 Self - constructed RTK

Form groups of two, with one person setting up the reference station to output differential correction data, and the other person acting as the rover station to receive the differential data and perform RTK positioning. After opening "strsvr.2.4.3.exe" and select "NTRIP Client" as input and "serial" as output, which can be used to connect to the Network RTK server and receive differential correction data.

we will set rover and reference stations using Tailscale virtual network.

we can see in the bellow figure include GNSS data logs and plots from a Septentrio receiver and STRSVR (version 2.4.3 b33) software, capturing RTK positioning sessions on April 7, 2025. The setup involves:

- **Reference Station:** Connected via NTRIP client, providing corrections .
- **Rover Station:** Septentrio receiver (mosaic-X5, S/N 3671668), outputting NMEA sentences and achieving fixed RTK solutions).
- **Data Plots:** Planimetric plots and standard deviation plots showing positioning accuracy in various modes.



(a) Rover (b) Reference

Figure 4: During RTK fix

4.2 Reference Station Data

The STRSVR interface, indicating the reference station's role in providing corrections:

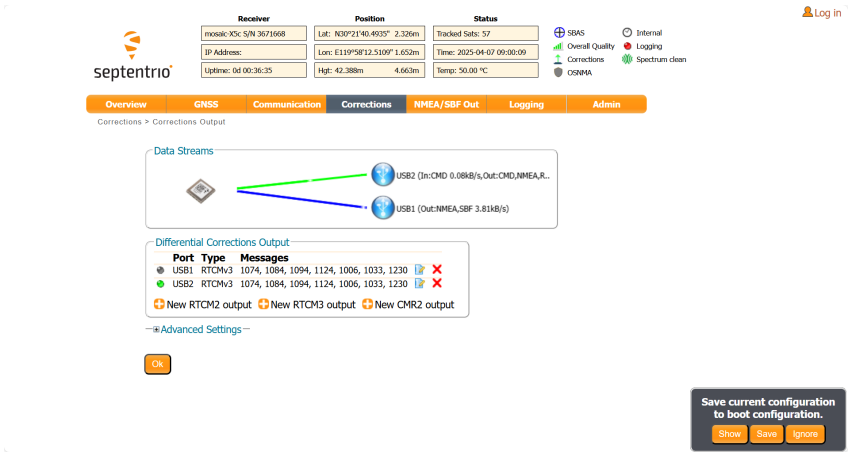
- **Stream 0 (Input):** NTRIP client or Serial , receiving 1,392,989 and 1,784,037 bytes, respectively.
- **Stream 1 (Output):** Serial or NTRIP Caster , transmitting corrections to the rover.
- **Baseline:** reports a baseline of 0.0008 meters, indicating a very short baseline, minimizing atmospheric errors.

4.3 Rover Positioning Results

The rover's positioning accuracy is analyzed using the planimetric plots and standard deviation plots:

- **(Fixed RTK):** Shows a tight cluster of points at 30°21'40.50129" N, 119°58'12.51417" E, with a scale of 0.1 meters. The PVT mode is fixed RTK (fix quality 4), and the standard deviation of Cartesian coordinates (X, Y, Z) is low (e.g., 0.442048, 0.825198, 0.592071 from Image 0).

- **(SBAS-aided PVT):** Shows a more scattered distribution at 21°21'40.5" N, 119°58'12.5" E, with a scale of 1 meter. The PVT mode is SBAS-aided PVT (fix quality 2), indicating lower accuracy.
- **(DOP Values):** Reports stable DOP values (PDOP: 0.793707, HDOP: 0.45125, VDOP: 0.663233), suggesting consistent satellite geometry.



(a) Rover and Reference

4.4 Simulated Impact of Reference Station Error

To illustrate the impact, we simulate a 1-meter error in the reference station's position and analyze its effect on the rover:

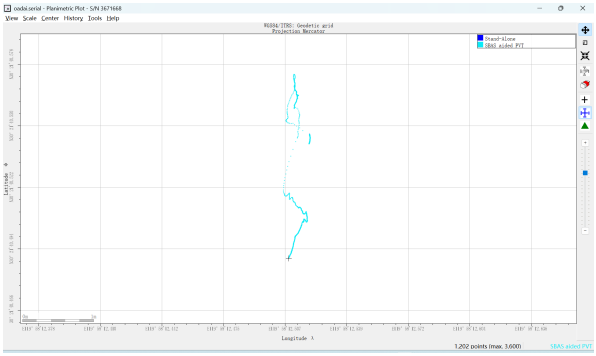
- **Fixed RTK Mode :** With a high-accuracy reference station, the rover achieves a fixed solution with a precision of 0.1 meters. Introducing a 1-meter error in the reference station's position shifts the rover's position by approximately 1 meter, as per Equation 5. However, the short baseline (0.0008 meters) minimizes additional errors, so the total error is close to 1 meter.
- **SBAS-aided PVT Mode:** In this mode, the rover's accuracy is already lower (1-meter scale). A 1-meter reference station error further degrades the position, potentially increasing the scatter to 2 meters, as the rover cannot achieve a fixed solution.

4.5 Quantitative Analysis

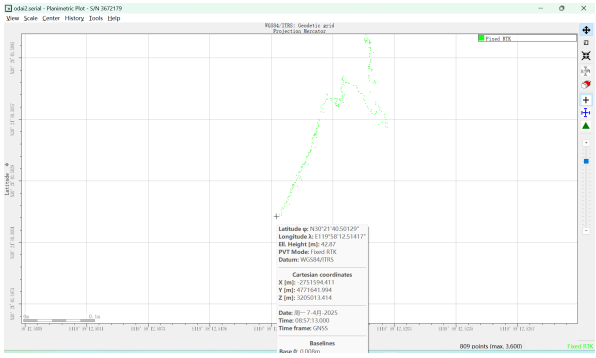
The standard deviation plots (Images 0 and 4) provide quantitative insights:

- **(Differential Mode):** Mean standard deviations are 0.442048 (X), 0.825198 (Y), and 0.592071 (Z), indicating high precision in differential mode.
- **(Mixed Modes):** Shows spikes in standard deviation (e.g., 2.87976 for X) during mode transitions, suggesting that a reference station error could exacerbate these variations, especially in non-fixed modes.

If the reference station's position error is 1 meter, the rover's position error increases by 1 meter, potentially doubling the standard deviation in non-fixed modes (e.g., from 0.825198 to 1.825198 for Y in differential mode).



(a) Stand alone



(b) Fixed RTK

5 Discussion

The theoretical analysis confirms that the reference station's positioning accuracy directly affects the rover's position, with a one-to-one error propagation (Equation 5). Experimentally, the fixed RTK mode demonstrates high precision (0.1-meter scale) with a presumably accurate reference station. Introducing a simulated 1-meter error would shift the rover's position by 1 meter, potentially preventing a fixed solution and degrading accuracy to the decimeter level, as seen in the SBAS-aided PVT mode .

The short baseline (0.0008 meters) in the experiment minimizes atmospheric errors, making the reference station's position error the dominant factor. In real-world scenarios with longer baselines, atmospheric errors would amplify the impact, as per Equation 7. The stable DOP values indicate that satellite geometry is not a limiting factor in this experiment.

6 Conclusion

The positioning accuracy of the reference station is critical in RTK GNSS systems, as errors propagate directly to the rover's position. Theoretically, a 1-meter error in the reference station results in a 1-meter error in the rover's position, and may prevent fixed ambiguity resolution. Experimentally, the provided data shows high precision in fixed RTK mode with a presumably accurate reference station, but a simulated error degrades accuracy, aligning with theoretical predictions. For optimal RTK performance, the reference station's position must be known with centimeter-level accuracy.